

ANDREW NOONAN

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EDUCATION

Vanderbilt University

Master of Science in Mechanical Engineering | GPA: 4.0/4.0

“Experimental Investigation of Venturi Microbubble Generation in a Viscous Fluid Using Temperature-Controlled Facility”

Nashville, TN

July 2025

Vanderbilt University

Bachelor of Engineering in Mechanical Engineering

Minor in Computer Science | GPA: 3.83/4.0 | Cum Laude

Nashville, TN

May 2023

WORK EXPERIENCE

Consultant Design Engineer

Vanderbilt University

Nashville, TN

July 2025 - Present

- Developing new experimental facilities for undergraduate laboratories

Graduate and Undergraduate Research Engineer

Vanderbilt Aerospace Design Laboratory (VADL)

Nashville, TN

May 2022 - July 2025

- Designed, fabricated, and characterized performance of novel Venturi air injectors for viscous fluids
- Designed VW Diesel engine test facility to measure lubricant flow rate and main gallery pressure
- Applied Fiber Bragg Gratings to structural health monitoring of composite airframes in collaboration with NASA AFRC
- Developed novel autorotation payload for entry, descent, and landing (EDL) ([Link](#))
- Designed, tested, and flight-validated RF payload detachment system for EDL Payload

Undergraduate Teaching Assistant

Vanderbilt University School of Engineering

Nashville, TN

August 2022 - December 2024

- Energetics Lab - Guided students through hands-on studies of Fluid Mechanics, Heat Transfer, and Thermodynamics with SI engine, wind tunnel, water tunnel, and heat exchanger labs
- FEA - CNC Machined and tested students' final projects, lectured, and guided students through assignments

Solutions Development Intern

SitelogIQ

Nashville, TN

May 2021 - August 2021

- Created customer-facing technical documents and project reports
- Wrote Python and Power Automate scripts to automate tasks and facilitate communication between teams

LEADERSHIP AND SERVICE EXPERIENCE

Team President, Graduate Mentor

NASA University Student Launch Initiative (USLI)

Nashville, TN

August 2022 - Present

- Led a 10-person engineering team in the design, fabrication, and launch of a high-powered rocket featuring a novel robotic payload for NASA's USLI competition (2022-2023); 3rd place nationally among 50+ teams ([Link](#))
- Led hands-on STEM educational activities to 1400+ K-12 students and community members

Member

AIAA Reusable Launch and Space Vehicle Technical Committee

Nashville, TN

December 2024 - Present

- Working with academia and industry professionals to develop an international student competition for AIAA

AWARDS AND HONORS

- **Leadership Awards:** Vanderbilt Mechanical Engineering Department Leadership Award, NASA Educational Engagement Award for most creative outreach activities
- **Michael Freeman Award:** Best paper award at AIAA Student Conference Region II ([Link](#))

SKILLS AND INTERESTS

- **Skills & Programming:** Fluid systems, CAD/CAM, MATLAB/Simulink, LabVIEW, Python, Ansys, Comsol
- **Interests:** Travel, live music, cooking, basketball, golf, stand-up comedy, running